Additional Problems: (q,r) Policy

1. An automotive warehouse stocks a variety of parts that are sold at neighborhood stores. The warehouse purchases a popular brand of oil filter for $1.50 each. It is estimated that the cost of order processing and receipt is $100 per order. The company uses an inventory carrying charge based on a 28 percent annual interest rate. The monthly demand for the filter follows a normal distribution with mean 280 and standard deviation 77. Order lead time is assumed to be five months. Assume that if a filter is demanded when the warehouse is out of stock, then the demand is back-ordered, and the cost assessed for each back-ordered demand is $12.80.

a. What are the optimal order quantity and the reorder level?

b. What is the expected safety stock level?

c. What is the expected average annual total cost?

SOLUTION:

a. *What are the optimal order quantity and the reorder level?*

We have the following:

- Given the distribution of monthly demand such that the mean demand per week is 280 units and there are 12 months in a year, we have $\lambda = (280)(12) = 3{,}360 \text{ units/year}$
- Given the distribution of monthly demand such that $X_D \sim N(280, 77^2)$ and a constant lead time of $\tau = 5$ months, we can derive the parameters for the demand during lead time as follows:

$$\mu_L = \tau\mu_D = (5)(280) = 1{,}400 \text{ units}$$

$$\sigma_L = \sqrt{\tau}\mu_D = \left(\sqrt{5}\right)(77) = 172.18 \text{ units}$$

Since the demand per month is normally distributed, so is the demand during lead time and we have $X_L \sim N(1{,}400, 172.18^2)$.

- For cost parameters, we have $b = \$12{:}80/\text{unit}$, $h = (0.28)(1.50) = \$0.42/\text{unit/year}$, and $K = \$100/\text{order}$.

Iteration $j = 0$

$$q_0 = \sqrt{\frac{2\lambda K}{h}} == \sqrt{\frac{2(3{,}360)(100)}{0.42}} = 1{,}264.91 \text{ units}$$

$$1 - F(r_0) = \frac{q_0 h}{b\lambda} \Rightarrow F(r_0) = 1 - \frac{q_0 h}{b\lambda} = 1 - \frac{(1{,}264.91)(0.42)}{(12.80)(3{,}360)} = 1 - 0.0124 = 0.9876$$

$$F(r_0) = 0.9876 \Rightarrow z_0 = 2.2445 \cong 2.24 \Rightarrow L(z_0) = 0.0044$$

$$n(r_0) = \sigma_L L(z_0) = (172.18)(0.0044) = 0.7576 \text{ units}$$

Iteration $j = 1$

$$q_1 = \sqrt{\frac{2\lambda[K + bn(r_0)]}{h}} = \sqrt{\frac{2(3{,}360)[100 + (12.80)(0.7576)]}{0.42}} = 1{,}324.82 \text{ units}$$

$$1 - F(r_1) = \frac{q_1 h}{b\lambda} \Rightarrow F(r_1) = 1 - \frac{q_1 h}{b\lambda} = 1 - \frac{(1{,}324.82)(0.42)}{(12.80)(3{,}360)} = 0.9871$$

$$F(r_1) = 0.9871 \Rightarrow z_1 = 2.2292 \cong 2.23 \Rightarrow L(z_1) = 0.0045$$

$$n(r_1) = \sigma_L L(z_1) = (172.18)(0.0045) = 0.7748 \text{ units}$$

Iteration $j = 2$

$$q_2 = \sqrt{\frac{2\lambda[K + bn(r_1)]}{h}} = \sqrt{\frac{2(3{,}360)[100 + (12.80)(0.7748)]}{0.42}} = 1{,}326.15 \text{ units}$$

$$1 - F(r_2) = \frac{q_2 h}{b\lambda} \Rightarrow F(r_2) = 1 - \frac{q_2 h}{b\lambda} = 1 - \frac{(1{,}326.15)(0.42)}{(12.80)(3{,}360)} = 0.9870$$

$$F(r_2) = 0.9871 \Rightarrow z_2 = 2.2262 \cong 2.23 \Rightarrow z_2 = z_1; \text{convergence!}$$

$$n(r_2) = n(r_1) = 0.7748 \text{ units}$$

Now, we have

$$q^* = q_2 = 1{,}326.15 \text{ units}$$

$$z^* = z_2 = 2.23$$

$$r^* = r_2 = \mu_L + z^*\sigma_L = 1{,}400 + (2.23)(172.18) = 1{,}783.96 \text{ units}$$

b. *What is the expected safety stock level?*

$$s^* = r^* - E[X_L] = r^* - \mu_L = 1783.96 - 1{,}400 = 383.96 \text{ units}$$

c. *What is the expected average annual total cost?*

$$\begin{aligned} C(q,r) &= c\lambda + \frac{K\lambda}{q} + h\left[\frac{q}{2} + s\right] + \frac{bn(r)\lambda}{q} \\ &= (1.50)(3{,}360) + \frac{(100)(3{,}360)}{1{,}326.15} + (0.42)\left[\frac{1{,}326.15}{2} + 383.96\right] \\ &\quad + \frac{(3)(0.7748)(3{,}360)}{1{,}326.15} = 5{,}040 + 439.75 + 253.37 + 25.14 \end{aligned}$$

2. Resolve Question 1 assuming that the excess demand is lost, and the unit shortage cost is \$18.80 per unit in this case.

SOLUTION:

a. *What are the optimal order quantity and the reorder level?*

We have the following:

- Given the distribution of monthly demand such that the mean demand per week is 280 units and there are 12 months in a year, we have $\lambda = (280)(12) = 3{,}360$ units/year
- Given the distribution of monthly demand such that $X_D \sim N(280, 77^2)$ and a constant lead time of $\tau = 5$ months, we can derive the parameters for the demand during lead time as follows:

$$\mu_L = \tau\mu_D = (5)(280) = 1{,}400\ units$$

$$\sigma_L = \sqrt{\tau}\mu_D = \left(\sqrt{5}\right)(77) = 172.18\ units$$

Since the demand per month is normally distributed, so is the demand during lead time and we have $X_L \sim N(1{,}400, 172.18^2)$.

- For cost parameters, we have $b = \$18.80/\text{unit}$, $h = (0.28)(1.50) = \$0.42/unit/year$, and $K = \$100/\text{order}$.

Iteration $j = 0$

$$q_0 = \sqrt{\frac{2\lambda K}{h}} == \sqrt{\frac{2(3{,}360)(100)}{0.42}} = 1{,}264.91 \text{ units}$$

$$1 - F(r_0) = \frac{q_0 h}{b\lambda + q_0 h} \Rightarrow F(r_0) = 1 - \frac{q_0 h}{b\lambda + q_0 h} = 1 - \frac{(1{,}264.91)(0.42)}{(12.80)(3{,}360) + (1{,}264.91)(0.42)} = 0.9917$$

$$F(r_0) = 0.9917 \Rightarrow z_0 = 2.3955 \cong 2.40 \Rightarrow L(z_0) = 0.0027$$

$$n(r_0) = \sigma_L L(z_0) = (172.18)(0.0027) = 0.4649 \text{ units}$$

Iteration $j = 1$

$$q_1 = \sqrt{\frac{2\lambda[K + bn(r_0)]}{h}} = \sqrt{\frac{2(3{,}360)[100 + (18.80)(0.4649)]}{0.42}} = 1{,}319.03 \text{ units}$$

$$1 - F(r_1) = \frac{q_1 h}{b\lambda + q_1 h} \Rightarrow F(r_1) = 1 - \frac{q_1 h}{b\lambda + q_1 h} = 1 - \frac{(1{,}319.03)(0.42)}{(18.80)(3{,}360) + (1{,}319.03)(0.42)} = 0.9913$$

$$F(r_1) = 0.9913 \Rightarrow z_1 = 2.3782 \cong 2.38 \Rightarrow L(z_1) = 0.0029$$

$$n(r_1) = \sigma_L L(z_1) = (172.18)(0.0029) = 0.4993 \text{ units}$$

Iteration $j = 2$

$$q_2 = \sqrt{\frac{2\lambda[K + bn(r_1)]}{h}} = \sqrt{\frac{2(3{,}360)[100 + (12.80)(0.4993)]}{0.42}} = 1{,}322.95 \text{ units}$$

$$1 - F(r_2) = \frac{q_2 h}{b\lambda + q_2 h} \Rightarrow F(r_2) = 1 - \frac{q_2 h}{b\lambda + q_2 h} = 1 - \frac{(1{,}322.95)(0.42)}{(18.80)(3{,}360) + (1{,}322.95)(0.42)} = 0.9913$$

$$F(r_2) = 0.9913 \Rightarrow z_2 = 2.3782 \cong 2.38 \Rightarrow z_2 = z_1; \text{convergence!}$$

$$n(r_2) = n(r_1) = 0.4993 \text{ units}$$

Now, we have

$q^* = q_2 = 1{,}322.95 \text{ units}$

$z^* = z_2 = 2.38$

$r^* = r_2 = \mu_L + z^*\sigma_L = 1{,}400 + (2.38)(172.18) = 1{,}809.99 \text{ units}$

b. *What is the expected safety stock level?*

$s^* = r^* - E[X_L] + n(r^*) = r^* - \mu_L + n(r^*) = 1809.99 - 1{,}400 + 0.4993 = 410.39 \text{ units}$

c. *What is the expected average annual total cost?*

$$\begin{aligned} C(q,r) &= c\lambda + \frac{K\lambda}{q} + h\left[\frac{q}{2} + s\right] + \frac{bn(r)\lambda}{q} \\ &= (1.50)(3{,}360) + \frac{(100)(3{,}360)}{1{,}322.95} + (0.42)\left[\frac{1{,}322.95}{2} + 410.39\right] \\ &+ \frac{(18.80)(0.4993)(3{,}360)}{1{,}322.95} = 5{,}040 + 449.98 + 253.98 + 23.84 \end{aligned}$$

3. Santori Bags sells a variety of briefcases and handbags. The bags are shipped from a manufacturer in Italy. During an interview, Marilyn Santori, the owner of the business, says that "In order to make sure that I never run out of stock, I try to keep at least three months' supply in stock. When the inventory drops to that level, I order another three months' supply. I have been using this method for the past sixteen years and it has worked for my company." Genuine leather bags cost the company $55 and are sold for $150 each. The cost of placing an order with the Italian manufacturer and receiving the bags is $750, and it takes five weeks to receive the shipment from Italy. The weekly demand is approximately normally distributed with a mean of 70 bags and a standard deviation of 20 bags. Marilyn's investments in the money market bring in approximately 12 percent earnings, and the insurance and tax related expenses amount to approximately 8 percent of the value of the goods in inventory.

a. What are the reorder level and order quantity for the bags that Marilyn is currently using?
b. What Type 2 service level is being achieved with the current policy?
c. What would the reorder level and order quantity be, if Marilyn were to use a Type 1 service level of 98 percent?
d. Suppose that the excess demand is back-ordered, and Marilyn offers a 10 percent discount in price to her customers when a customer order has to be back-ordered. In this case, determine the optimal values of the reorder level and order quantity.

SOLUTION:

a. *What are the reorder level and order quantity for the bags that Marilyn is currently using?*

$$\begin{aligned} r &= (3)(4)(70) = 840 \; units \\ q &= (3)(4)(70) = 840 \; units \end{aligned}$$

b. *What Type 2 service level is being achieved with the current policy*?

We first need to determine the parameters of the distribution of demand during lead time. Given the distribution of demand $X_D \sim N(70, 20^2)$ and a constant lead time of $\tau = 5\ weeks$, we can derive the parameters for the demand during lead time as follows:

$$\mu_L = (5)(70) = 350\ units$$

$$\sigma_L = \sqrt{(5)(20)^2} = 44.72\ units$$

Since the demand is normally distributed, so is the demand during lead time and we have $X_L \sim N(350, 44.72^2)$.

To determine β, we need to determine $n(r)$. To this end, let us consider $L(z)$ first. We have

$$L(z) = L\left(\frac{840 - 350}{44.72}\right) = L(10.96) \approx 0$$

Hence, we have

$$n(r) = \sigma_L L(z) = (44.72)(0) = 0\ units$$

Consequently,

$$1 - \beta = \frac{n(r)}{q} \Rightarrow 1 - \beta = \frac{0}{840} \Rightarrow 1 - \beta = 0$$
$$\beta = 1$$

That is, a 100% Type 2 service level is being achieved by the current policy

c.

A Type 1 service level of 98 percent implies $\alpha = 0.98$. Then, using a 48-week year, we have

$$F(r) = 0.98 \Rightarrow z_{\alpha=0.98} = 2.0538$$

$$r = 350 + (2.0538)(44.72) = 441.85\ units$$

$$q = \sqrt{\frac{(2)(750)(48)(70)}{(0.12 + 0.08)(55)}} = 676.89\ units$$

d.

The discount offer can be used to quantify the cost associated with back-logging. That is, for modeling purposes we can use $b = (0.10)(150) = \$15/unit$. Using a 48-week year, we have $\lambda = (48)(70) = 3,360\ units/yr$. Unit inventory holding cost is $h = (0.12 + 0.08)(55) = \$11/unit/yr$.

Iteration for j = 0: We have the following:

$$q_0 = \sqrt{\left(\frac{(2)(750)(3,360)}{11}\right)} = 676.89\ units$$

$$1 - F(r_0) = \frac{(676.89)(11)}{(15)(3{,}360)} = 0.1477$$
$$F(r_0) = 0.8523 \Rightarrow z_0 = 1.0462\ (using\ Excel) \Rightarrow z_0 \approx 1.05 \Rightarrow L(z_0) = 0.0757\ (using\ table)$$
$$n(r_0) = (44.72)(0.0757) = 3.3853\ units$$

Iteration for j = 1: We have the following:

$$q_1 = \sqrt{\frac{(2)(3,360)[(750) + (15)(3.3853)]}{11}} = 699.43\ units$$
$$1 - F(r_1) = \frac{(699.43)(11)}{(15)(3{,}360)} = 0.1527$$
$$F(r_1) = 0.8473 \Rightarrow z_1 = 1.0249\ (using\ Excel) \Rightarrow z_1 \approx 1.03 \Rightarrow L(z_1) = 0.0787\ (using\ table)$$
$$n(r_1) = (44.72)(0.0757) = 3.5195\ units$$

Iteration for j = 2: We have the following:

$$q_2 = \sqrt{\frac{(2)(3,360)[(750) + (15)(3.5195)]}{11}} = 700.75\ units$$
$$1 - F(r_2) = \frac{(700.75)(11)}{(15)(3,360)} = 0.1529$$
$$F(r_2) = 0.8471 \Rightarrow z_1 = 1.0241\ (using\ Excel) \Rightarrow z_1 \approx 1.02 \Rightarrow L(z_1) = 0.0802\ (using\ table)$$
$$n(r_2) = (44.72)(0.0802) = 3.5865\ units$$

$$n(r_2) = n(r_1)$$

We have converged to the optimal solution. We have

$$q^* = q_2 = 700.75\ units$$
$$r^* = r_2 = 350 + (1.02)(44.72) = 395.61\ units$$

4. A company uses 85 circuit boards a day in a particular assembly cell. The cell manager of the process places orders for the boards according to the following rule: Order when the amount on hand drops to 625 boards. Orders are delivered approximately six days after being placed. The delivery time is normally distributed with a mean of 6 days and a standard deviation of 1.1 days.

What is the probability that the supply of circuit boards will be exhausted before the order is received, if boards were reordered when the amount on hand drops to 625 boards?

SOLUTION:

Given the distribution of lead time in days such that $X_\tau \sim N(6, 1.1\ 2\ )$ and a constant daily demand of $\mu_D = 85\frac{units}{day}$, we can derive the parameters for the demand during lead time as follows:

$$\mu_L = (6)(85) = 510\ units$$
$$\sigma_L = \sqrt{(85)^2(1.1)^2} = 93.5\ units$$

Since the lead time is normally distributed, so is the demand during lead time and we have $X_L \sim N(510, 93.5^2)$.

We have $r = 625\ units$. Then, we have

$$F(r) = \Phi\left(\frac{625 - 510}{93.5}\right) = \Phi(1.2299) = 0.8907$$

which implies a probability of a stock-out during lead time of